Many-Task Federated Fine-Tuning via Unified Task Vectors

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01 Context & approach

The challenge

When clients hold multiple tasks, local updates can interfere within and across clients. Maintaining a separate model for every task increases storage and communication. *MaTU* transfers one unified task vector per client with lightweight adjustments that preserve task identity.

Our contribution

Represent multiple tasks with one vector, binary task masks and scalar rescalers.
Combine same-task updates and transfer information between tasks with aligned signs.
Avoid retaining client-specific model weights at the server across rounds.

1 Unify local tasks

Fine-tune LoRA parameters per task. Elect a shared sign and the largest aligned magnitude for each vector entry.

2 Preserve task identity

Attach a binary mask and scalar rescaler per task; transmit them with the unified vector.

3 Share related updates

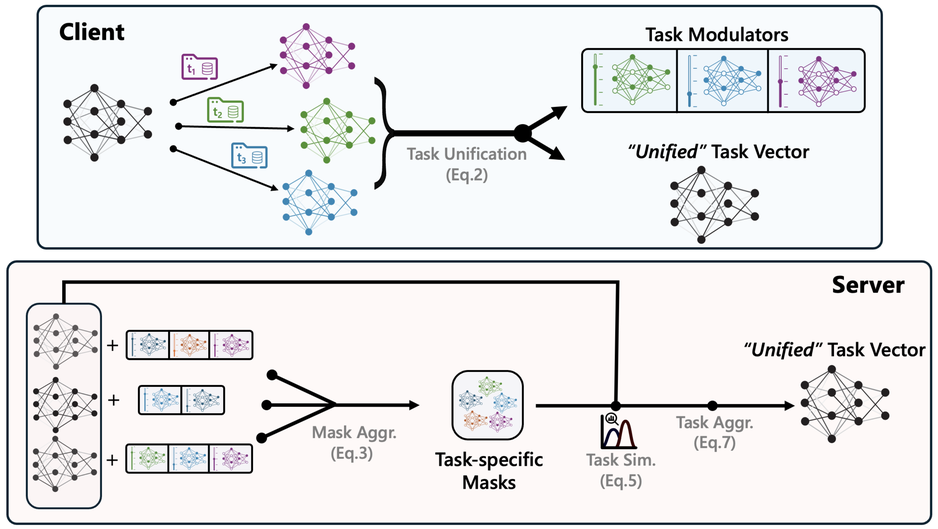
Aggregate same-task updates, combine sign-aligned tasks, and return unified vectors with updated modulators for the next round.

Scope

Vision classification with ViT-B/32 and LoRA; other modalities and backbones remain untested. No uncertainty reported; the eight-task MaT-FL baseline uses 300 rounds versus 100 for *MaTU*.

02 Method

Unify client task vectors while retaining task-specific directions and magnitudes.



Core formulation

$$\tau = \sigma \odot \mu$$

τ: unified task vector; σ: sign of the summed task vectors; μ: largest magnitude agreeing with that sign, entry by entry.

$$\dot{\tau}_t = \lambda_t m_t \odot \tau, \qquad \theta_t = \theta_p + \dot{\tau}_t$$

Mask m_t keeps task-aligned entries; scalar λ_t rescales them. Add the recovered task vector to pretrained weights θ_p.

Paper

Code

03 Results

79.47%
average task accuracy
8 tasks; multiple tasks per client.

8.04 Mbit/task
communication per round
MaT-FL: 20.43 Mbit/task/round.

Multiple tasks per client: accuracy and communication

Method	Avg. accuracy (%)	Mbit/task/round
Individual centralized	90.21	—
FedAvg	51.63	20.43
FedPer	48.37	17.16
MaT-FL	67.60	20.43
MaTU (Ours)	**79.47**	**8.04**

8 vision tasks; ViT-B/32, LoRA rank 16. 30 clients, 1 local epoch, task concentration ζt=0.5, 20% participation. 100 rounds; MaT-FL: 300. No uncertainty reported.

Scaling to the 30-task benchmark

Method	Normalized accuracy (%)
MaT-FL	52.62
MaTU (Ours)	**77.40**

30 tasks; accuracy normalized to individual task fine-tuning. ViT-B/32, 30 clients, 300 rounds, 1 local epoch, task concentration ζt=0.2, full participation.

Takeaway

MaTU reaches **79.47% accuracy at 8.04 Mbit/task/round** across eight tasks, exceeding MaT-FL with 100 versus 300 training rounds. Across 30 tasks, normalized accuracy rises from 52.62% to 77.40%, while remaining below individual task fine-tuning.